

Week 3: Fundamentals of Probability and Statistics I

PLSC 309 Quantitative Political Analysis

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Office Hours Schedule Change

- **Mondays, 1:00–2:30 p.m.**, starting from next week.

Final Project: Checkpoint 1

Due: next Wednesday, September 16 at 11:59 p.m. (midnight)

Instructions: Posted on Canvas

Question 6

Countries with higher levels of trade openness are more likely to join international organizations.

- IV: Level of trade openness/Trade openness
- DV: Joining IOs/IO membership/Likelihood of joining IOs
- Direction: Higher IV $\rightarrow$ greater likelihood of DV

A hypothesis usually contains an implicit or explicit comparison.

Remember: The IV and DV are the concepts that vary, not their particular values or the predicted direction

Feedback: Writing a Hypothesis

A hypothesis is usually **a single sentence** stating the predicted association or causal relationship between an independent variable (IV) and a dependent variable (DV).

Some standard examples:

- [Cases with a higher/lower level of X] are more/less likely to experience Y.
- As X increases, Y increases/decreases.
- X has a positive/negative effect on Y.
- [Cases with X] are more/less likely to experience Y than [cases without X].
- [Change (increase or decrease) in X] increases/decreases Y.

...

What Did We Learn Last Time?

- Random variable
- Probability distribution
- Probability distribution functions (PMF, PDF, CDF)
- Expected value
- Variance and standard deviation
- Covariance and correlation

Review: Random Variable

Random Variable

A **random variable** is a **function** $X : \Omega \rightarrow \mathbb{R}$ that assigns a real number to each outcome in the sample space.

- The *input* is an outcome $\omega \in \Omega$ (which outcome occurs)
- The *output* is a number $X(\omega) \in \mathbb{R}$ (the numerical value)

The “**randomness**” comes from not knowing which ω will occur. (Think about the pollster example)

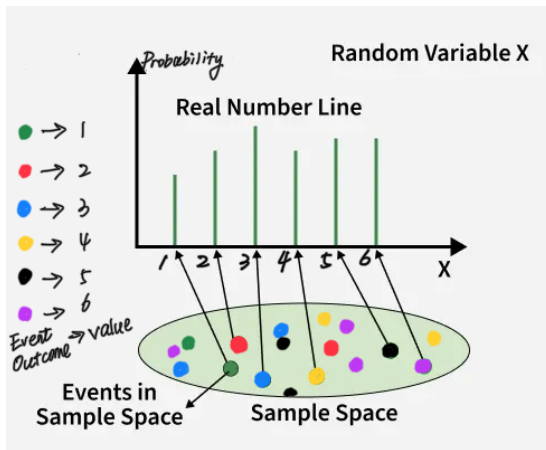
Non-random variable (fixed, deterministic, constant):

- e.g., π (pi), the speed of light in a vacuum (a universal physical constant),...

Review: Probability Distribution

Probability Distribution

A **probability distribution** describes how the possible values of a random variable are distributed along with their chances of occurring.



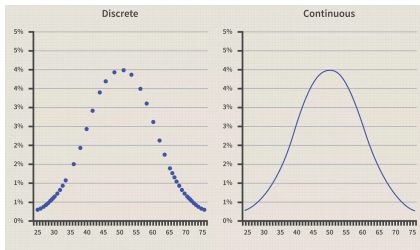
Review: Probability Mass Function vs. Probability Density Function

PMF

For a discrete random variable X :

$$p_X(x) = P(X = x)$$

The exact probability for each value

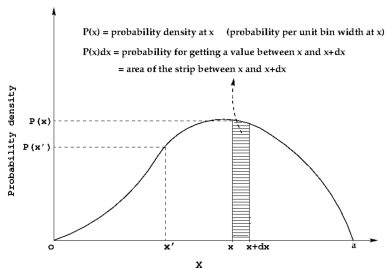


PDF

For a continuous random variable X :

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

The area under the curve between two specific points



Review: Expected Value

A **single number** used to summarize **the center** of a probability distribution.

Expected Value

The **expected value** (or **mean**) of a random variable X is:

$$\mathbb{E}[X] = \begin{cases} \sum x \cdot p(x), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx, & \text{if } X \text{ is continuous} \end{cases}$$

Interpretation: The expected value is a **weighted average** of possible values, where the weights are the probabilities (or densities).

Notation: $\mathbb{E}[X]$, $\bar{X}$, μ , and μ_X all denote the same thing.

Review: Variance

A **single number** used to describe **the spread** of possible values around the expected value.

Variance

The **variance** of a random variable X is:

$$\text{Var}(X) = \mathbb{E} [(X - \mathbb{E}[X])^2]$$

- Deviations above and below the mean ($X - \mathbb{E}[X]$) would **cancel out**
- Squaring makes all deviations **non-negative**
- Larger deviations get **penalized more heavily**
- A single number: take the average of the **squared deviation from the mean**.

Review: Standard Deviation

Variance has a problem for interpretation because it uses **squared units** given that its calculation involves squaring the differences between each data point and the mean.

Standard deviation: **A single number** used to describe **the spread** of possible values around the expected value in its original unit.

Standard Deviation

The **standard deviation** of X is the square root of its variance:

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Review: Covariance

A **single number** used to quantify how two variables vary together.

Covariance

The **covariance** between two random variables X and Y is:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

It is the expected value of the product of their deviations from their respective means.

$X - \mathbb{E}[X]$	$Y - \mathbb{E}[Y]$	Product
+	+	+
-	-	+
+	-	-
-	+	-

Review: Correlation

A problem with covariance is that **its magnitude depends on the units of measurement**.

Correlation: **a single number** between -1 and 1 that shows **the strength and direction** of a relationship between two variables

Correlation

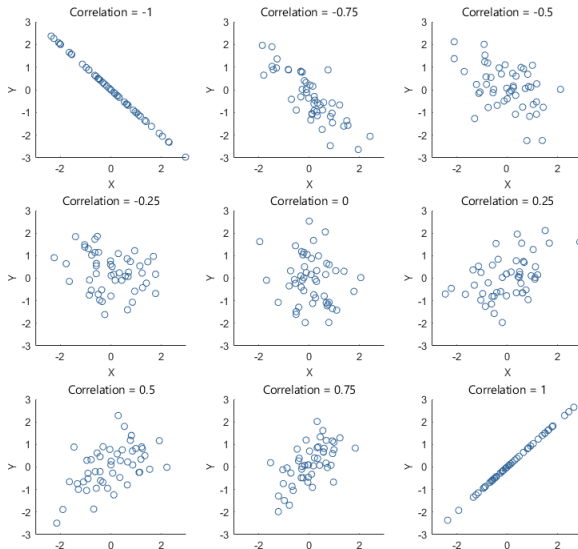
The **correlation** between X and Y is:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Correlation	Interpretation
$\rho = 1$	Perfect positive linear relationship
$\rho \approx 0$	Little or no linear relationship
$\rho = -1$	Perfect negative linear relationship

Correlation Coefficient

Realizations of couples of random variables X and Y
with different correlation coefficients



Today's Agenda

- Sampling and Estimation (learning population from samples)
- Properties of Estimators
- Interval Estimation

Acknowledgement: Some slides in today's lecture are adapted from Scott Cunningham's [GOV 2001 course materials at Harvard](#) and from Brandon Stewart's [SOC 500 course materials at Princeton](#).

Part 1. Sampling and Estimation

The Practical Problem

You are an analyst for a candidate running for office in Pennsylvania.

Your boss asks you a simple question:

What percentage of Pennsylvania voters support our candidate?

Why does this matter?

- The campaign wants to assess its **chance of winning**.
- Your boss wants a **single number**:

Support = ?%

But there is a practical problem:

You cannot ask every voter in Pennsylvania.

Access, time, and cost make surveying the entire voting population practically impossible.

So What Can We Do?

We want to know:

Support among **all** Pennsylvania voters

But we can only observe:

Support among **some** Pennsylvania voters

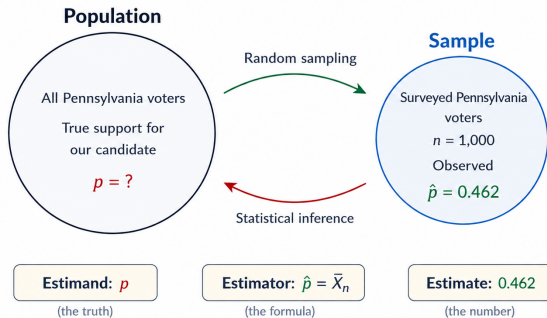
So we:

- 1 Select a **random sample** of voters.
- 2 Measure support in that random sample.
- 3 Use the sample to **estimate** support in the population.

Population → **Sample** → **Estimate**

Learning from Random Samples

From Population to Sample



Statistical inference is the process of using data from a (small) random sample to draw conclusions, test claims, or estimate unknown properties of a (large) population.

From Finite Population to Random Sample

A finite population of N units with values $x_1, \dots, x_N$.

Population mean:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i$$

Population variance:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

Random sampling: Draw n units at random. Each draw X_i has the same distribution (**identically distributed**) and draws do not affect each other (**independent**).

I.I.D.: $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} F$

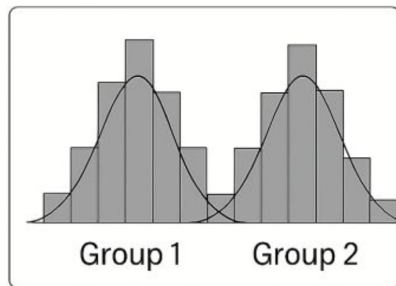
① **Independent:**

$$X_i \perp X_j \quad \text{for all } i \neq j$$

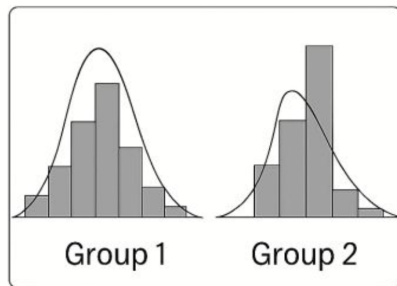
② **Identically distributed:** Each X_i has the same probability distribution.

Note: x_i denotes the fixed value of unit i in the population, whereas X_i denotes the random value obtained on the i -th draw from the population. Thus, x_i is fixed, while X_i is a random variable.

Assumptions met



Different shapes



IID: Independent and Identically Distributed —

Formalizing the Framework

Precise definitions for the visual we just saw

Estimand $\theta = T(F)$:

- The specific target quantity or population parameter that you want to calculate or estimate in a statistical study.
- Examples: $\mu = \mathbb{E}[X]$, $\sigma^2 = \text{Var}(X)$

Estimator $\hat{\theta}_n = h(X_1, \dots, X_n)$:

- A function of the sample—a **random variable** (it has a distribution!)

Estimate:

- A realized value of the estimator—a *number*, not a random variable

Common mistake: “My estimator was 0.462.” No—your *estimate* was 0.462. The estimator is the formula.

Estimand, Estimator, and Estimate (in plain language)

In our Pennsylvania polling example:

p
Estimand

$\hat{p} = \bar{X}_n$
Estimator

0.462
Estimate

Estimand

The population quantity
we want to know.

p

Fixed but unknown

Estimator

The rule we use to
estimate the estimand
from a sample.

$\hat{p} = \bar{X}_n$

Random variable

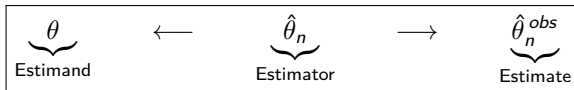
Estimate

The value produced by
the estimator in this
sample.

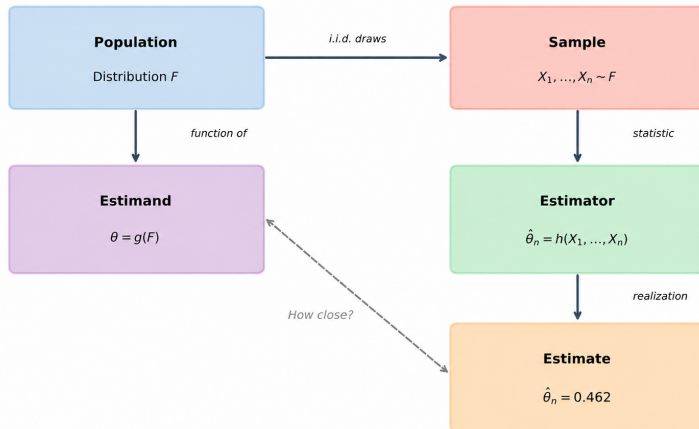
0.462

A realized number

General notation:



The Estimation Framework



The Sample Mean as an Estimator

Natural idea: Estimate μ with the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Key insight: $\bar{X}_n$ is itself a *random variable*.

- Different samples give different values of $\bar{X}_n$
- $\bar{X}_n$ has its own distribution—the **sampling distribution**
- We want to understand this distribution

In our Pennsylvania polling example: $\bar{X}_n = 0.462$ is one draw from the sampling distribution of the sample mean.

Statistics is about understanding how estimators behave across repeated samples.

The Three Distributions: What Are They?

① **Population Distribution** $F(x)$:

The true distribution of values in the population—aka. the **data-generating process (DGP)**.

② **Empirical Distribution**:

The distribution we observe in the sample $X_1, \dots, X_n$ —what we actually have. For our polling example, this is a series of 1s and 0s.

③ **Sampling Distribution** $G(\hat{\theta}_n)$:

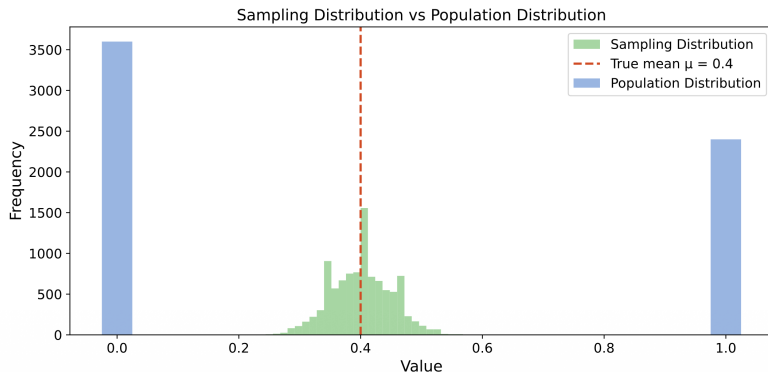
The distribution of the **estimator** over repeated samples.

Our observed sample mean, $\bar{X}_n = 0.462$, is *one draw* from this distribution.

Trick question: “The sampling distribution is the distribution of θ .” True or false?

False—it is the distribution of $\hat{\theta}_n$, the estimator.

Sampling Distribution vs. Population Distribution



Population is binary (0s and 1s). But the sampling distribution of $\bar{X}_n$ is bell-shaped and concentrated around the true mean.

Simulating One Poll in R

Suppose the true support for our candidate is:

$$p = 0.40$$

and we survey $n = 1,000$ Pennsylvania voters.

```
set.seed(123)
p_true <- 0.40 # true parameter (the estimand)
n <- 1000 # sample size

# Simulate n=1000 voters: 1 = support, 0 = not support
support <- rbinom(n, size = 1, prob = p_true)
# Look at the first six simulated voters
head(support)
```

```
## [1] 0 1 0 1 1 0
```

```
# Calculate the sample mean
mean(support)
```

```
## [1] 0.393
```

What If We Repeat the Poll?

Now suppose we conduct the same poll 1000 times.

```
R <- 1000 # 1000 draws (random sampling)
```

```
estimates <- replicate(  
  R,  
  mean(rbinom(n, size = 1, prob = p_true))  
)
```

```
head(estimates) # the first six sample means
```

```
## [1] 0.386 0.417 0.409 0.384 0.396 0.390
```

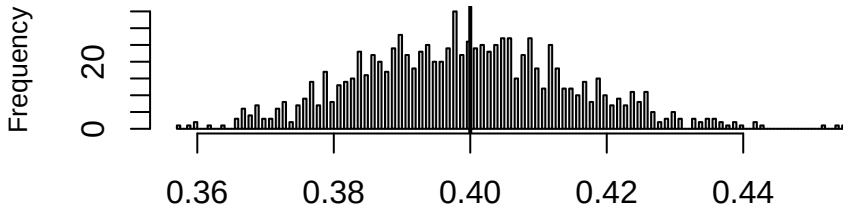
```
mean(estimates) # expected value of the sample mean
```

```
## [1] 0.399405
```

The Sampling Distribution in R

```
hist(  
  estimates,  
  breaks = 150,  
  main = "Sampling Distribution of the Sample Mean",  
  xlab = "Estimated Support",  
  cex.main = 0.8, cex.lab = 0.8, # customize label sizes  
)  
  
abline(v = p_true, lwd = 2)
```

Sampling Distribution of the Sample Mean



Part 2. Properties of Estimators

Many Estimators for the Same Estimand

Suppose our goal is to estimate the **population mean**:

$$\mu = \mathbb{E}[X]$$

There are many possible estimators we could construct from a sample $X_1, \dots, X_n$.

Sample mean

$$\hat{\mu}_1 = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Uses every observation equally.

First observation

$$\hat{\mu}_2 = X_1$$

Uses only one observation.

Weighted mean

$$\hat{\mu}_3 = \frac{2}{3} \bar{X}_n + \frac{1}{3} X_1$$

Gives some observations more weight.

Sample median

$$\hat{\mu}_4 = \text{Median}(X_1, \dots, X_n)$$

Uses the middle of the sample.

All are functions of the sample. But are they equally good estimators of μ ?

Which one should we choose?

- Gets the **right answer on average** $\rightarrow$ *Unbiasedness*
- Does not vary too much from sample to sample $\rightarrow$ *Precision / Efficiency*
- Gets closer to the true value as n increases $\rightarrow$ *Consistency*
- Has a sampling distribution that becomes predictable for large n $\rightarrow$ *Asymptotic behavior*

Properties of Estimators

The properties of an estimator describe the characteristics of the sampling distribution.

Finite-Sample Properties (apply for any sample size):

- **Unbiasedness**: Is the estimator centered at the true value? $\mathbb{E}[\hat{\mu}] = \mu$
- **Efficiency**: Does the estimator have relatively small variance?
 $\text{Var}(\hat{\mu}_1) < \text{Var}(\hat{\mu}_2)$

Asymptotic Properties (apply when n is large):

- **Consistency**: Does the estimator converge to the true parameter value as n grows? $\hat{\theta}_n \xrightarrow{P} \theta$
- **Asymptotic Normality**: Does the sampling distribution approach a normal distribution as n grows?

Definition

$$\text{Bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta$$

$\mathbb{E}[\hat{\theta}_n]$ is the expected value of the sample mean estimator.

θ is the estimand.

Unbiased: $\text{Bias}(\hat{\theta}_n) = 0$, i.e., $\mathbb{E}[\hat{\theta}_n] = \theta$.

Is the sample mean unbiased?

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \cdot n\mu = \mu$$

Yes! On average (over repeated samples), $\bar{X}_n$ hits the true value.

Variance of the Estimator

Sampling variance: How spread out is $\hat{\theta}_n$ around its mean?

$$\text{Var}(\hat{\theta}_n) = \mathbb{E} \left[\left(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] \right)^2 \right]$$

For the sample mean:

$$\text{Var}(\bar{X}_n) = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right) = \frac{\sigma^2}{n}$$

Key insight: Variance shrinks as n increases—more data means more precision.

FYI: Why Is $\text{Var}(\bar{X}_n) = \sigma^2/n$?

Variance of the Sample Mean $\bar{X}_n$

Result: If $X_1, X_2, \dots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$, then

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Derivation:

①	$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$	Start from the definition of the sample mean $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$
②	$= \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right)$	Use: $\text{Var}(aY) = a^2 \text{Var}(Y)$ (a is a constant)
③	$= \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) \right]$	Use the general formula: $\text{Var}\left(\sum X_i\right) = \sum \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$
④	$= \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} 0 \right]$	Independence: $\text{Cov}(X_i, X_j) = 0$ for $i \neq j$
⑤	$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$	The second term is 0
⑥	$= \frac{1}{n^2} \sum_{i=1}^n \sigma^2$	Identically distributed: $\text{Var}(X_i) = \sigma^2$ for all i
⑦	$= \frac{1}{n^2} (n\sigma^2) = \frac{\sigma^2}{n}$	There are n terms in the sum

Key Insight: The variance of the sample mean decreases at rate $1/n$.
More data $\Rightarrow$ smaller variance $\Rightarrow$ more precise estimator!

Standard Error: Precision in Practice

Standard Error: The standard deviation of the estimator.

$$SE(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)}$$

For the sample mean: $SE(\bar{X}_n) = \sigma/\sqrt{n}$

How fast does precision improve?

- $n = 100$: $SE = \sigma/10$
- $n = 10,000$: $SE = \sigma/100$

To cut SE in half, you need **4× the sample size** because $\sqrt{4n} = 2\sqrt{n}$.

Estimating the Standard Error

But in practice, the population standard deviation σ is **unknown**.

We estimate σ using the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}$$

Then, the **estimated standard error** is

$$\widehat{SE}(\bar{X}_n) = \frac{s}{\sqrt{n}}$$

From a single sample, we can calculate both:

$$\bar{X}_n = \text{Point Estimate} \qquad \frac{s}{\sqrt{n}} = \text{Estimated Standard Error}$$

Notation Cheat Sheet

Population

N

Population size

F

Population distribution

$$\mu = \mathbb{E}[X]$$

Population mean

$$\sigma^2 = \text{Var}(X)$$

Population variance

σ

Population standard deviation

p

Population proportion

Sample

n

Sample size

$X_1, \dots, X_n$

Random sample

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

Sample mean

s^2

Sample variance

s

Sample standard deviation

$\hat{p}$

Sample proportion

Estimation

θ

Estimand (true value)

$\hat{\theta}_n$

Estimator (random variable)

$\hat{\theta}_n^{obs}$

Estimate (realized value)

$\text{Var}(\hat{\theta}_n)$

Sampling variance

$SE(\hat{\theta}_n)$

Standard error

Mean Squared Error (MSE)

Definition

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E} [(\hat{\theta}_n - \theta)^2]$$

MSE measures how far the estimator is from the true parameter, on average.

Why does $\text{MSE} = \text{Bias}^2 + \text{Variance}$?

Add and subtract $\mathbb{E}[\hat{\theta}_n]$:

$$\begin{aligned}\text{MSE}(\hat{\theta}_n) &= \mathbb{E} [(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] + \mathbb{E}[\hat{\theta}_n] - \theta)^2] \\ &= \underbrace{\mathbb{E} [(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2]}_{\text{Var}(\hat{\theta}_n)} + \underbrace{(\mathbb{E}[\hat{\theta}_n] - \theta)^2}_{\text{Bias}(\hat{\theta}_n)^2}\end{aligned}$$

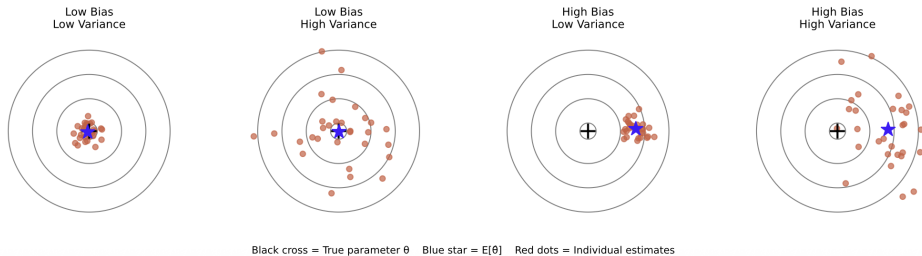
Therefore,

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + \text{Bias}(\hat{\theta}_n)^2$$

If the estimator is unbiased: $\text{Bias}(\hat{\theta}_n) = 0$, so $\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n)$.

The Bias-Variance Trade-Off

Which matters more—bias or variance? Neither alone. We care about the overall accuracy of the estimator (MSE).



Central Limit Theorem: The Intuition

Suppose we repeatedly draw random samples of size n from a population with mean μ and variance σ^2 .

Each sample produces a different sample mean:

$$\bar{X}_1, \bar{X}_2, \bar{X}_3, \dots$$

The **Central Limit Theorem (CLT)** tells us that, as n becomes large, the sampling distribution of $\bar{X}$ becomes approximately **normal**:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Even if the population itself is not normally distributed!

- Center: $\mathbb{E}[\bar{X}] = \mu$
- Spread: $SE(\bar{X}) = \sigma/\sqrt{n}$
- Shape: approximately normal for sufficiently large n

Why Does the Central Limit Theorem Matter?

The CLT gives us an approximate sampling distribution for the sample mean:

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Standardizing:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$$

This allows us to use known probabilities from the standard normal distribution. Therefore, we can quantify the uncertainty around our estimate.

CLT $\longrightarrow$ Normal Sampling Distribution $\longrightarrow$ Confidence Intervals

Part 3. Interval Estimation

From Point Estimation to Interval Estimation

Recall our Pennsylvania polling example:

$$\hat{p} = 0.462$$

This is our **point estimate** of the true population support p .

But should we believe that

$$p = 0.462 \quad \text{exactly?}$$

From Point Estimation to Interval Estimation

Recall our Pennsylvania polling example:

$$\hat{p} = 0.462$$

This is our **point estimate** of the true population support p .

But should we believe that

$$p = 0.462 \quad \text{exactly?}$$

No. A different random sample would give a different $\hat{p}$.

The point estimate alone does not tell us how **uncertain** our estimate is.

The Goal

Instead of reporting a single number, construct a **range of plausible values** for the unknown population parameter.

What Is an Interval Estimate?

A **point estimate** gives us one best guess: $\hat{\theta}_n$.

An **interval estimate** gives us a range around that estimate:

$$\text{Point Estimate} \pm \text{Margin of Error}$$

For many estimators:

$$\hat{\theta}_n \pm \underbrace{z^*}_{\text{Critical Value}} \times \underbrace{SE(\hat{\theta}_n)}_{\text{Standard Error}}$$

Two ingredients determine the width:

- **Standard error:** How much does the estimator vary across samples?
- **Critical value z^* :** How far we move from the point estimate?

From the Sampling Distribution to an Interval

Suppose the sampling distribution is approximately normal:

$$\hat{\theta}_n \approx N\left(\theta, SE(\hat{\theta}_n)^2\right)$$

Standardizing gives:

$$Z = \frac{\hat{\theta}_n - \theta}{SE(\hat{\theta}_n)} \approx N(0, 1)$$

For a 95% confidence interval, we want the middle 95% of this distribution.

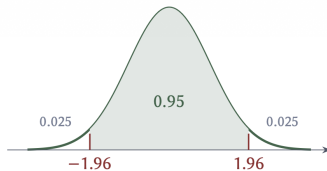
$$P(-1.96 \leq Z \leq 1.96) \approx 0.95$$

The remaining 5% is split equally between the two tails:

$$2.5\% \quad | \quad \mathbf{95\%} \quad | \quad 2.5\%$$

Why 1.96?

For a 95% confidence interval, we want to capture the **middle 95%** of a standard normal distribution.



Since $\hat{Z}_n \approx N(0, 1)$ by the CLT, this 95% area applies to our quantity directly:

$$\mathbb{P}\left(-1.96 < \frac{\bar{Y} - \mu}{\hat{\sigma}/\sqrt{n}} < 1.96\right) \approx 0.95$$

Note: A **normal distribution** is a symmetric, bell-shaped distribution characterized by its mean and standard deviation. The **standard normal distribution** is a normal distribution with mean 0 and standard deviation 1. We use Z to denote a random variable following this distribution: $Z \sim N(0, 1)$.

Different Confidence Levels

Different confidence levels require different critical values:

Confidence Level	Critical Value	Interval
90%	1.645	$\hat{\theta} \pm 1.645SE$
95%	1.960	$\hat{\theta} \pm 1.960SE$
99%	2.576	$\hat{\theta} \pm 2.576SE$

Higher confidence $\Rightarrow$ larger critical value $\Rightarrow$ wider interval

Trade-off: More confidence requires less precision.

Example: Pennsylvania Poll

Suppose we survey $n = 1,000$ Pennsylvania voters:

$$\hat{p} = 0.462$$

For a sample proportion:

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.462(0.538)}{1000}} \approx 0.0158$$

The 95% confidence interval is:

$$\hat{p} \pm 1.96SE(\hat{p})$$

$$0.462 \pm 1.96(0.0158)$$

$$[0.431, 0.493]$$

Confidence Intervals from Repeated Samples

Suppose the true proportion of Pennsylvania voters who support the candidate is

$$p = 0.40.$$

We repeatedly draw random samples of $n = 1,000$ voters from this population.

For each sample, we calculate:

Point Estimate $\longrightarrow$ Standard Error $\longrightarrow$ 95% Confidence Interval.

We repeat this process **100 times**.

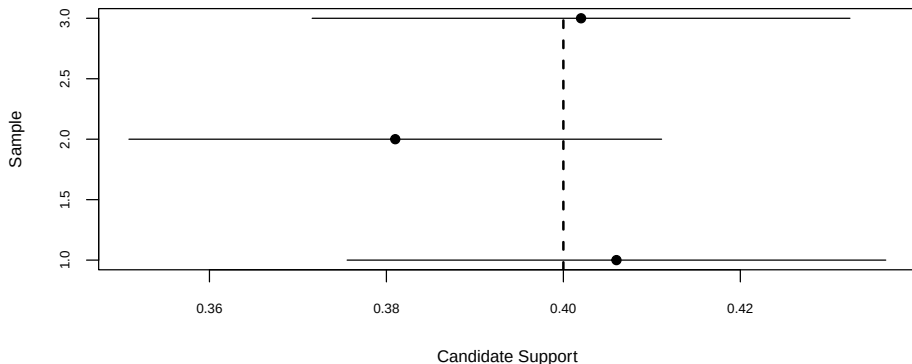
Confidence Intervals from Repeated Samples

Let's start with just **three random samples**.

Each dot represents the **point estimate** $\hat{p}$, and each horizontal line represents its **95% confidence interval**.

The dashed vertical line represents the **true population value**, $p = 0.40$.

95% Confidence Intervals from Repeated Samples (3 times)

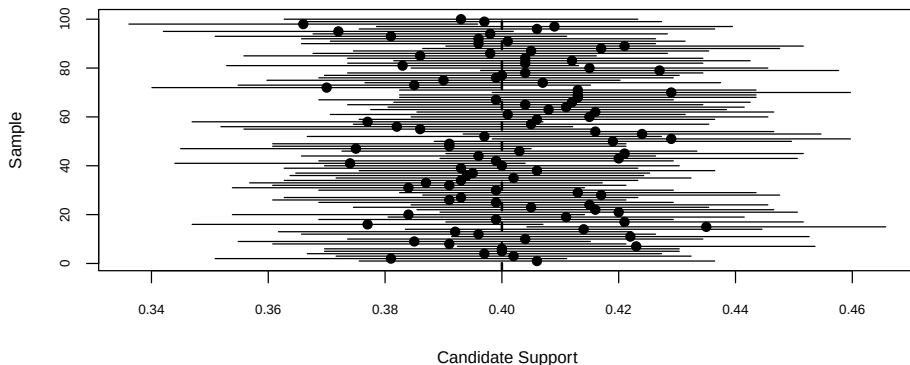


What If We Repeat This 100 Times?

Now imagine repeating exactly the same procedure **100 times**.

Each random sample produces a different point estimate and therefore a different confidence interval.

95% Confidence Intervals from Repeated Samples (100 times)



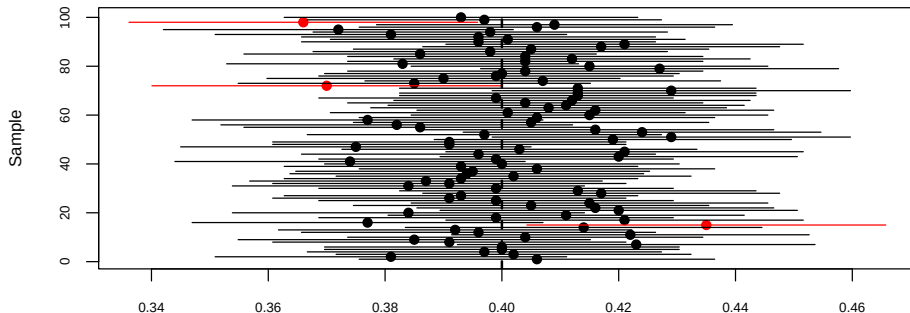
Do All Confidence Intervals Contain the Truth?

No. Some confidence intervals **miss the true population value**.

In the figure below:

- Black intervals **contain** the true value $p = 0.40$.
- Red intervals **do not contain** the true value.

95% Confidence Intervals from Repeated Samples (100 times)



How to Interpret a 95% Confidence Interval

Correct

- In repeated sampling, **95% of confidence intervals constructed this way contain the true parameter μ .**

Incorrect

- “There is a 95% probability that μ lies within this interval.”
- “95% of the observations in the population lie within this interval.”
- “95% of sample means will lie within this interval.”

Why? μ is **fixed**. Across repeated samples, the confidence interval changes—not μ .

Key Takeaways

- Statistical inference
- Random sampling (iid.)
- Estimand, estimator, estimate
- Sampling distribution
- Properties of estimators
- Sampling variance, standard error
- Confidence interval and its interpretation